SmartTherm Technical Specifications

Document AT-SPEC-3200, Revision C, January 2024

Covers: SmartTherm Lite (AT-2100) and SmartTherm Pro (AT-3200).

1. Product overview

SmartTherm is a line of connected thermostats for residential HVAC systems.
Built-in motion sensor detects when people are home.
Both models can be controlled at the device or through the Acme Home mobile app.

2. Model comparison

Connectivity, SmartTherm Lite (AT-2100): Wi-Fi 802.11 b/g/n, 2.4 GHz only.
Connectivity, SmartTherm Pro (AT-3200): Wi-Fi 802.11 b/g/n/ac, dual-band 2.4 / 5 GHz, plus Bluetooth 5.0 LE.
Occupancy sensing, SmartTherm Lite (AT-2100): not supported.
Occupancy sensing, SmartTherm Pro (AT-3200): passive infrared (PIR) sensor.
Remote temperature sensors, SmartTherm Lite (AT-2100): not supported.
Remote temperature sensors, SmartTherm Pro (AT-3200): up to 6 SmartSensor units.
Display, both models: 3.5 in. colour touchscreen.

3. Hardware: SmartTherm Pro (AT-3200)

3.1 Wireless communication module

The AT-3200 integrates an Espressif ESP32-S3 module providing Wi-Fi and Bluetooth LE radios.
The module maintains a persistent TLS 1.3 connection to the Acme Cloud service.
Setpoint commands from the Acme Home app are received over Wi-Fi and applied within 2 seconds.

3.2 Occupancy sensor

The AT-3200 includes a Panasonic EKMC-series passive infrared (PIR) motion sensor.
Detection range: up to 5 m. Horizontal field of view: 94 degrees.
Occupancy state (occupied / unoccupied) is sampled every 30 seconds and reported to the scheduling engine.
After 20 minutes with no detected motion, the device enters Away mode and applies the eco setpoint.

3.3 Temperature sensing

Onboard sensor: Sensirion SHT40 digital temperature and humidity sensor, accuracy ±0.2 °C.
Remote sensing: up to 6 SmartSensor remote units, each with its own temperature sensor.
SmartSensor units are sold separately and communicate over a 915 MHz sub-GHz radio link.
Readings from all paired sensors are averaged, or weighted by occupancy when Follow Me is enabled.

4. Software and processing

4.1 Auto-Schedule

Auto-Schedule builds a weekly heating and cooling schedule from the user's manual adjustments. Algorithm details are described in Acme engineering whitepaper AT-WP-07.

4.2 Firmware

Shipping firmware for the AT-3200: version 5.4.2 (as of January 2024).

4.3 Processor

Application processor: ARM Cortex-M4F at 120 MHz, 1 MB flash, 256 KB SRAM.

Specifications subject to change without notice.